Matter is anything that has mass and occupies space.

All living and non living things are made of matter.

Matter can exist as solid,liquid and gas.

Substances can exist as different states of matter under different temperature and pressure conditions.

Key Words

- **Mass** → Amount of matter in an object
- **Volume** → Space occupied by an object

Examples of Matter

Air, Water, Book, Pencil, Human body

Not Matter

Light, Sound, Heat.

Properties of Matter

- Matter has mass.
- Matter occupies space (volume).
- Matter is made of tiny particles.
- Matter has inertia (resists changes in motion).
- Matter can change its state when heated or cooled.

Particle Nature of Matter

Everything around us is made up of tiny particles.

Kinetic Particle Theory-The differences in the properties of states of matter can be explained through this theory.

Particle arrangement + forces of attraction + particle movement ⟶ determine the properties of matter.

The **Kinetic Particle Theory** states that:

1. All matter is made of tiny particles.
2. Particles are always moving (motion)
3. There are spaces between particles.
4. Particles attract each other.
5. Heating increases particle movement.

Brownian Motion

Brownian Motion is the random movement of tiny particles in a liquid or gas. This happens because the particles of the liquid or gas are always moving and bump into them.

Example

- Pollen grains moving randomly in water.
- Dust particles moving in a beam of sunlight.

Importance

Brownian motion provides evidence that particles of matter are in constant motion.

States of Matter

Matter exists in different forms called **states of matter**.

Main States

1. Solid
2. Liquid
3. Gas

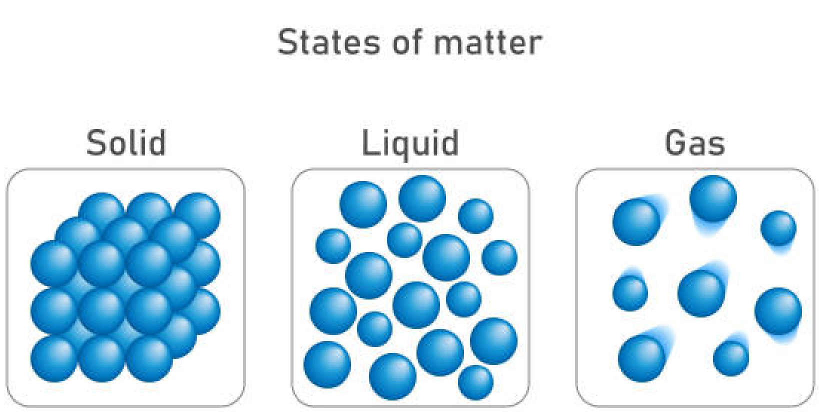
Other States

4. Plasma-High-energy state of matter,Consists of charged particles (ions)-sun, stars, lightning.

5. Bose-Einstein Condensate (BEC)-Extremely low-temperature state

   Atoms behave as a single quantum entity

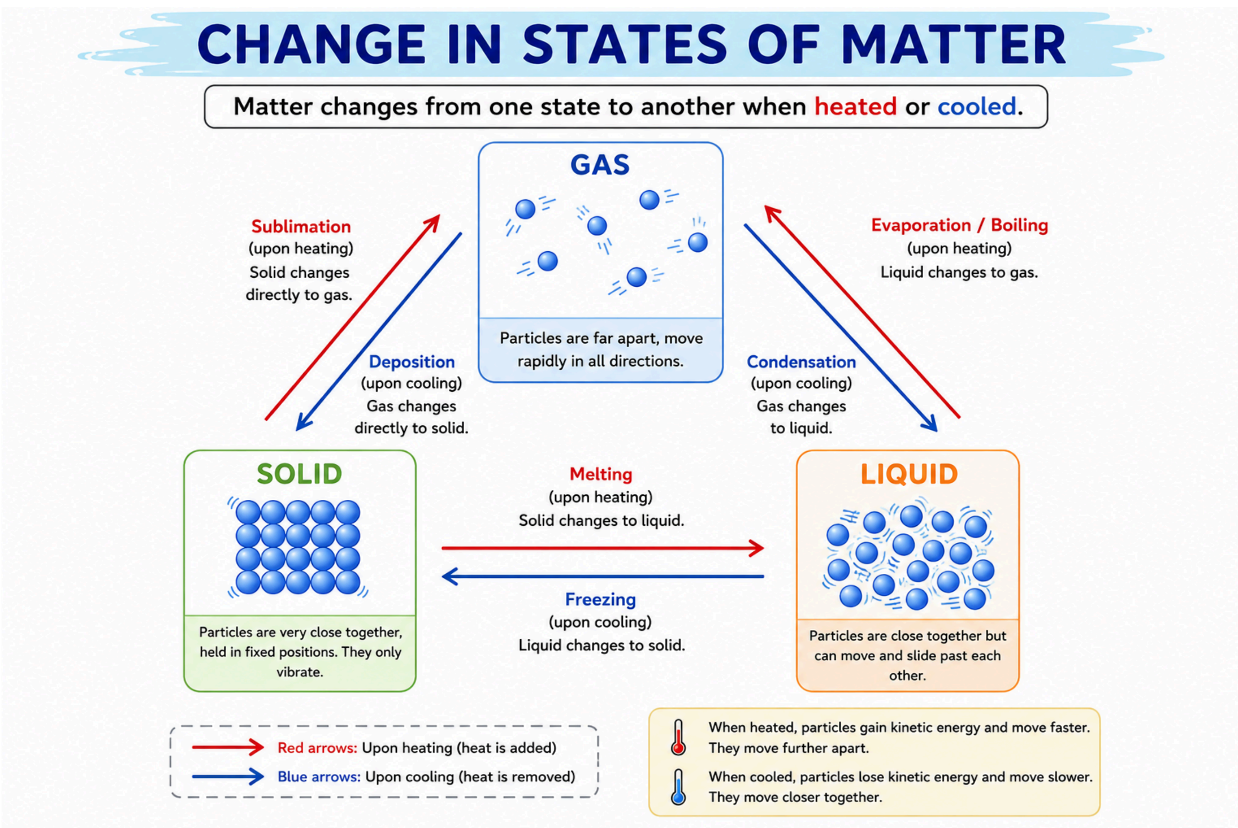
Solids, Liquids and Gases

	Solids	**Liquids**	**Gases**
Shape	fixed	No fixed shaped	No fixed shape
Volume	Fixed	fixed	No fixed volume
compressibility	Cannot be compressed	Cannot be compressed	Can be compressed
Particle arrangement	Closely packed	Close together	Far apart
Particle movement	Vibrate only	Slide past each other	Move freely
Diagram			

Why do solids, liquids and gases behave differently?"

The properties of solids, liquids and gases depend on:

- Arrangement of particles
- Movement of particles
- Force of attraction between particles



Matter changes from one state to another when heated or cooled

Melting	**Freezing**	**Boiling**	**Evaporation**	**Condensation**
Solid ⟶ Liquid	Liquid ⟶ Solid	Liquid ⟶ Gas	Liquid ⟶ Gas	Gas ⟶ Liquid

Example: Ice ⟶ Water	Example: Water ⟶ Ice	Example: Water ⟶ Steam	Example: Wet clothes drying Occurs at temperatures below the boiling point.	Example: Water droplets on a cold bottle
Melting Point The temperature at which a solid melts. Ice ⟶ Water	**Freezing Point** The temperature at which a liquid freezes. Water ⟶ Ice	**Boiling Point** The temperature at which a liquid boils. Water ⟶ Steam		

Everyday Applications

Refrigerators use evaporation and condensation.
Air conditioners cool by evaporation.
Water purification uses evaporation.
Dry ice is used in transporting vaccines.
Sublimation is used in mothballs.
Condensation forms clouds.
Diffusion helps oxygen enter the blood.

Boiling vs Evaporation

Boiling	**Evaporation**
Occurs at boiling point	Occurs at any temperature below boiling point
Happens throughout liquid	Happens only at surface
Rapid process	Slow process
Bubbles form	No bubbles

Sublimation

Some solids change directly into gases without becoming liquids.

Examples:

- Dry ice
- Camphor
- Naphthalene balls
- Iodine crystals

Reverse process:
Deposition (gas → solid)-Frost formation

Diffusion

Diffusion is the movement of particles from a region of high concentration to a region of low concentration.

Examples

- Perfume smell spreading

- Incense stick smell spreading
- Food colouring spreading in water

Important

Diffusion is fastest in gases because gas particles move very quickly.

Rate of Diffusion

Gas > Liquid > Solid

Factors Affecting Diffusion

- Temperature
- Particle size
- State of matter

Higher temperature → Faster diffusion

Smaller particles → Faster diffusion

Temperature and Particle Movement

When a substance is heated:

- Particles gain kinetic energy.
- Particles move faster.
- Particles move further apart.

When a substance is cooled:

- Particles lose kinetic energy.
- Particles move slower.
- Particles come closer together.

ATOMS, ATOMIC STRUCTURE AND SUB-ATOMIC PARTICLES

Atomic Structure is the branch of chemistry that studies the structure and composition of atoms.

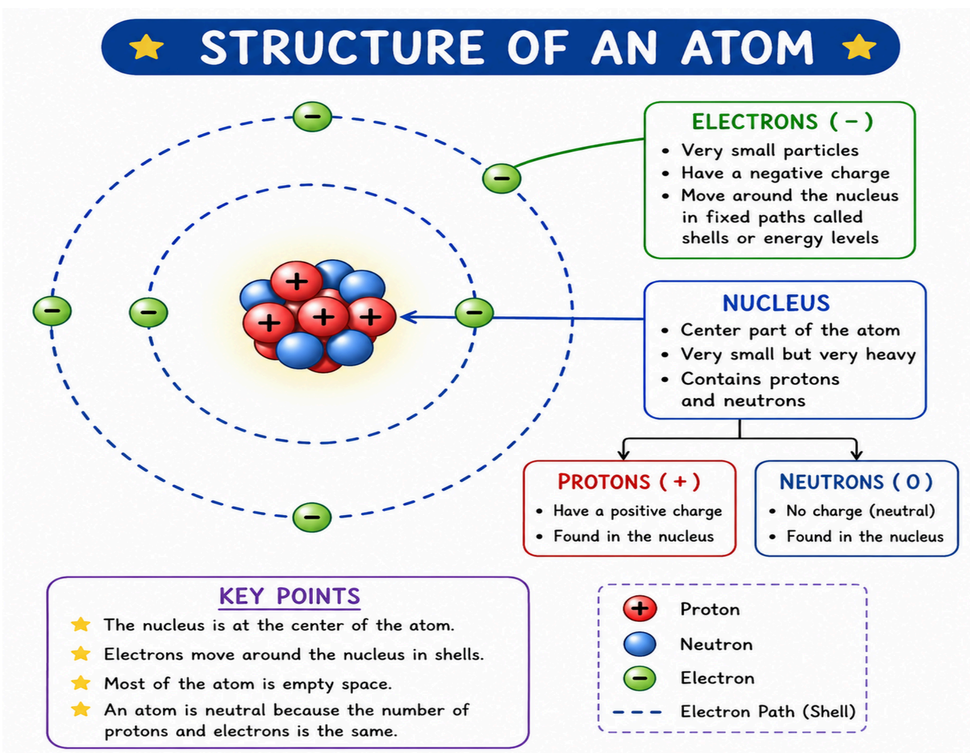
What is an Atom?

An atom is the smallest unit of an element that retains its chemical properties.

All atoms consist of:

- Nucleus
  - Protons (+)
  - Neutrons (0)
- Electrons (-) moving around the nucleus

Structure of an Atom



A. Nucleus

The nucleus is the central part of the atom.

Contains:

- Protons (+)
- Neutrons (0)

B. Electrons

- Negatively charged particles
- Move around the nucleus in energy levels (shells)

Important Facts

Most of the atom's mass is concentrated in the nucleus.

Most of the atom is empty space. Electrons occupy the space around the nucleus.

PHYSICAL AND CHEMICAL CHANGES

Physical and chemical changes are two fundamental ways matter can be altered. A physical change affects the form or appearance of a substance without changing its chemical composition, while a chemical change involves a rearrangement of atoms and the formation of new substances with different properties.

Physical Changes:

- **Definition:**

A physical change alters the form or appearance of a substance, but not its chemical makeup.

- **Examples:**

Melting ice, boiling water, crushing a can, dissolving sugar in water, cutting wood.

- **Reversibility:**

Many physical changes are reversible (e.g., freezing water back into ice).

- **Key Characteristics:**

Changes in state (solid, liquid, gas), shape, size, or texture are typical of physical changes.

Chemical Changes:

1. **Definition:**

A chemical change, also known as a chemical reaction, involves the formation of new substances with different chemical compositions and properties.

2. **Examples:**

Burning wood, rusting iron, cooking food, digestion.

3. **Reversibility:**

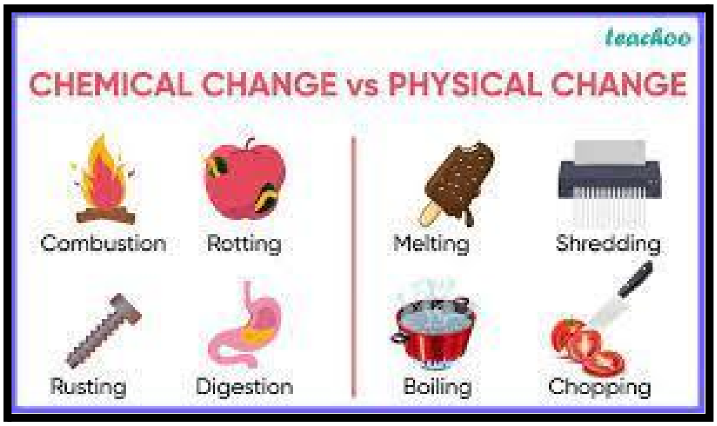
Chemical changes are often difficult or impossible to reverse without another chemical reaction.

4. **Key Characteristics:**

Often involve a change in color, temperature, the release or absorption of energy, and the formation of a precipitate or gas.

In simpler terms: If you can change something back to its original form without a chemical reaction, it's likely a physical change. If a new substance with different properties is formed, it's a chemical change.

Examples of chemical changes would be burning, cooking, rusting, and rotting. Examples of physical changes could be boiling, melting, freezing, and shredding.



Physical change	Chemical change
Physical changes are mostly reversible.	Chemical changes are not reversible.
No new substances are formed.	One or more new substances are formed.
The substance retain its chemical properties.	the new substances formed have different properties from the original substance.

Law of Conservation of Mass

During a physical or chemical change,

Mass is neither created nor destroyed.

The total mass of reactants equals the total mass of products.

Example:

Iron + Oxygen → Rust

Total mass remains the same in a closed system.

Examples of Physical Changes

Examples:

- Melting butter
- Breaking glass
- Cutting paper
- Dissolving sugar
- Freezing water

Examples of Chemical Changes

Examples:

- Rusting of iron
- Burning wood
- Cooking food
- Digestion of food
- Milk turning into curd